

From jun-2022 | Page 17:

Q5 (d): Calculate the magnetic flux density of the Earth's magnetic field. Use the equation $F = B \times I \times l$

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Q5 (b): Describe the forces acting on the upper magnet. Use the idea of magnetic fields in your answer.

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Q5 (c) (i) 1: State the direction of the force on the wire.

Q5 (c) (i) 2: State the direction of the force on the magnet.

Q5 (c) (ii): Calculate the length of the wire in the magnetic field. Use an equation selected from the list of equations given at the end of the question paper.

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Q10 (c): Compare how the device operates when used as a loudspeaker with how the device operates when used as a microphone.

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Q5 (b) (i): Draw an arrow on Figure 14 to show the direction of the force, F , acting on the wire.

Q5 (b) (ii): State one practical way of reversing the direction of force F .

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Q5 (b) (iii): Calculate the magnitude of the magnetic flux density between the two poles of the magnet.

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Q5 (b) (i): Calculate the value of K . State the unit.

Q5 (b) (ii): The magnets are held the same distance apart but with the N-pole of one magnet now facing the S-pole of the other magnet. The value of K does not change. State how the force would compare with the force in part (i).

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Q6 (c): Describe how the student could develop the investigation to determine if the size of the force between the magnet and the coil is directly proportional to the size of the current in the coil.

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Q6 (d): The student reverses the direction of the current in the coil. Describe how the student can bring the system back into equilibrium without making any changes to the magnet.

From specimen | Page 22:

Q6 (c) (i): Explain why the train acts in this way by considering the forces involved.

From specimen | Page 24:

Q7 (a): Calculate the current in the wire. Use an equation from the formula sheet. Give your answer to an appropriate number of significant figures.

From specimen | Page 26:

Q7 (b): Give three ways the student should develop his investigation to improve the quality of his results.

From specimen | Page 27:

Q7 (d): A power transmission wire hangs so that it is at right angles to the Earth's magnetic field. Although this magnetic field is constant, the cable experiences a changing force. Explain why the force experienced by the cable changes.